SIMULATION EXPERIMENT – 5

Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

CO Mapping: CO2
Bloom's Level: K5 – Evaluate
WK Level: WK4 – Design of Coupler/Divider

Objectives of this Experiment

1. To design microwave directional coupler and two-way power divider structures operating at 10 GHz.
2. To determine coupling factor, isolation, insertion loss, and directivity using S-parameters.
3. To analyze the power division ratio, port matching, and output-port isolation.
4. To compare the original design with a modified design after changing the coupling-section length by 5%.

Software Required: MATLAB / CST Studio Suite / Ansys HFSS / ADS

Objective linkage: Design the coupler and divider, analyze their S-parameters, calculate performance parameters, and compare the results with theoretical values.

Experimental Procedure

A. Directional Coupler: Design a 10 GHz directional coupler with 10 dB coupling and 50 Ω ports. Sweep the frequency from 8–12 GHz. Obtain S11, S21, S31, and S41. Calculate coupling, isolation, insertion loss, and directivity.

B. Two-Way Power Divider: Design a two-way equal power divider operating at 10 GHz. Analyze S11, S21, S31, and S23. Verify whether both output ports receive approximately equal power, corresponding to −3.01 dB per output for an ideal divider.

C. Optimization: Increase the coupling-section length by 5% and repeat the analysis. Compare coupling, isolation, insertion loss, and directivity before and after modification.

INPUT PARAMETER TABLE – VALUES TO BE ENTERED

S.No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
1	Operating frequency	f_0	10	GHz
2	Reference impedance	Z_0	50	Ω
3	Frequency sweep	$f_{\text{start}} - f_{\text{stop}}$	8–12	GHz
4	Coupling target	C	10	dB
5	Divider output	Power division	1:1	Ratio
6	Ideal divider output	S_{21}, S_{31}	−3.01 each	dB

S.No.	Input Parameter	Symbol / Setting	Value to Enter	Unit
7	Initial coupling-section length	L_c	Design-dependent	mm
8	Modified length	L_c'	$1.05L_c$	mm
9	Coupler port configuration	—	Four-port	—
10	Divider port configuration	—	Three-port	—

FORMULAS TO BE USED

1. Coupling factor

$$C = -20\log_{10} | S_{31} | \text{ dB}$$

2. Isolation

$$I = -20\log_{10} | S_{41} | \text{ dB}$$

3. Insertion loss

$$IL = -20\log_{10} | S_{21} | \text{ dB}$$

4. Directivity

$$D = I - C$$

5. Power division ratio

$$\frac{P_2}{P_3} = \frac{| S_{21} |^2}{| S_{31} |^2}$$

For an ideal equal power divider:

$$P_2 = P_3 = \frac{P_{\text{in}}}{2}$$
$$10\log_{10}(0.5) = -3.01 \text{ dB}$$

6. Percentage deviation

$$\% \text{ Deviation} = \frac{| \text{Simulated} - \text{Theoretical} |}{| \text{Theoretical} |} \times 100$$

KEY CONDITIONS AND THRESHOLD VALUES

Parameter	Expected value	Condition
Operating frequency	10 GHz	Within 8–12 GHz sweep
Reference impedance	50 Ω	For all ports
Coupling factor	10 dB	Deviation ≤ 5%
Isolation	≥ 25 dB	Higher is better
Insertion loss	≤ 1.5 dB	Lower is better
Directivity	≥ 15 dB	Higher is better
Divider output 1	Approximately −3.01 dB	Equal power split
Divider output 2	Approximately −3.01 dB	Equal power split
Power ratio	1:1	Equal output power
Optimization	5% length increase	Compare before and after

Note: These are design targets. Actual performance depends on the coupler/divider geometry, substrate, and electromagnetic simulation results.

MATLAB CODE

The following code displays **all six graphs in one figure layout** and prints the calculated performance parameters.

The curves are an illustrative analytical model, not actual CST/HFSS electromagnetic results. Replace the model data with measured or full-wave simulated S-parameters for validated results.

```
clc;

clear;

close all;


% SIMULATION EXPERIMENT 5

% Directional Coupler and Two-Way Power Divider


% INPUT PARAMETERS

f0 = 10e9;

Z0 = 50;

f = linspace(8e9,12e9,401);

fg = f/1e9;
```

```
x = (f-f0)/f0;

% DIRECTIONAL COUPLER - ORIGINAL DESIGN
C1 = 10 + 12*x.^2;
IL1 = 0.5 + 2*x.^2;
I1 = 30 - 20*x.^2;
D1 = I1-C1;

S11 = 0.04 + 0.03*abs(x);
S21 = 10.^(-IL1/20);
S31 = 10.^(-C1/20);
S41 = 10.^(-I1/20);

% MODIFIED COUPLER - APPROXIMATE 5% LENGTH CHANGE
C2 = 10 + 12*x.^2 - 0.4;
IL2 = 0.45 + 2*x.^2;
I2 = 32 - 20*x.^2;
D2 = I2-C2;

S11m = 0.03 + 0.025*abs(x);
S21m = 10.^(-IL2/20);
S31m = 10.^(-C2/20);
S41m = 10.^(-I2/20);

% TWO-WAY POWER DIVIDER
S11d = 0.04 + 0.03*abs(x);
S21d = 10.^(-3.1/20)*ones(size(f));
S31d = 10.^(-3.0/20)*ones(size(f));
S23d = 10.^(-25/20)*ones(size(f));

% CREATE ONE FIGURE WITH SIX GRAPHS
figure('Color','w','Position',[50 50 1400 800]);
```

```
% 1. COUPLER S-PARAMETERS
subplot(2,3,1);
plot(fg,20*log10(abs(S11)),'LineWidth',1.8);
hold on;
plot(fg,20*log10(abs(S21)),'LineWidth',1.8);
plot(fg,20*log10(abs(S31)),'LineWidth',1.8);
plot(fg,20*log10(abs(S41)),'LineWidth',1.8);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Directional Coupler S-Parameters');
legend('S11','S21','S31','S41','Location','best');

% 2. COUPLING FACTOR
subplot(2,3,2);
plot(fg,C1,'LineWidth',2);
hold on;
plot(fg,C2,'--','LineWidth',2);
yline(10,':k');
grid on;
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');
title('Coupling Factor');
legend('Original','Modified','10 dB Target');

% 3. ISOLATION
subplot(2,3,3);
plot(fg,I1,'LineWidth',2);
hold on;
plot(fg,I2,'--','LineWidth',2);
yline(25,':k');
```

```
grid on;
xlabel('Frequency (GHz)');
ylabel('Isolation (dB)');
title('Isolation');
legend('Original','Modified','25 dB Minimum');

% 4. INSERTION LOSS
subplot(2,3,4);
plot(fg,IL1,'LineWidth',2);
hold on;
plot(fg,IL2,'--','LineWidth',2);
yline(1.5,':k');
grid on;
xlabel('Frequency (GHz)');
ylabel('Insertion Loss (dB)');
title('Insertion Loss');
legend('Original','Modified','1.5 dB Maximum');

% 5. DIRECTIVITY
subplot(2,3,5);
plot(fg,D1,'LineWidth',2);
hold on;
plot(fg,D2,'--','LineWidth',2);
yline(15,':k');
grid on;
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');
title('Directivity');
legend('Original','Modified','15 dB Minimum');

% 6. POWER DIVIDER
subplot(2,3,6);
```

```
plot(fg,20*log10(abs(S11d)),'LineWidth',1.8);
hold on;
plot(fg,20*log10(abs(S21d)),'LineWidth',1.8);
plot(fg,20*log10(abs(S31d)),'LineWidth',1.8);
yline(-3.0103,':k');
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Two-Way Power Divider');
legend('S11','S21','S31','Ideal -3.01 dB',...
    'Location','best');

sgtitle('SIMULATION EXPERIMENT 5: COUPLER AND POWER DIVIDER',...
    'FontSize',14,'FontWeight','bold');

% CALCULATE VALUES AT 10 GHz
[~,idx] = min(abs(f-f0));

C0 = C1(idx);
I0 = I1(idx);
IL0 = IL1(idx);
D0 = D1(idx);

Cmod = C2(idx);
Imod = I2(idx);
ILmod = IL2(idx);
Dmod = D2(idx);

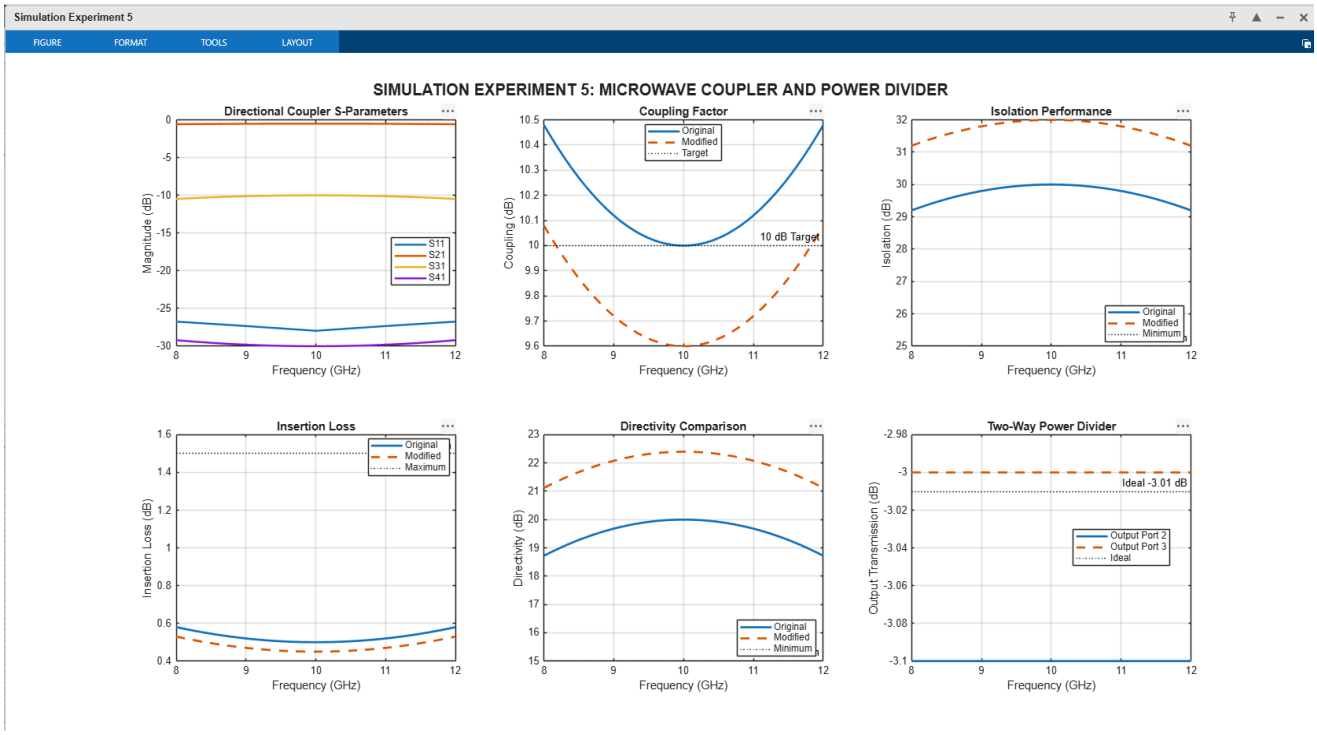
P2 = 20*log10(abs(S21d(idx)));
P3 = 20*log10(abs(S31d(idx)));
IsoDivider = -20*log10(abs(S23d(idx)));
```

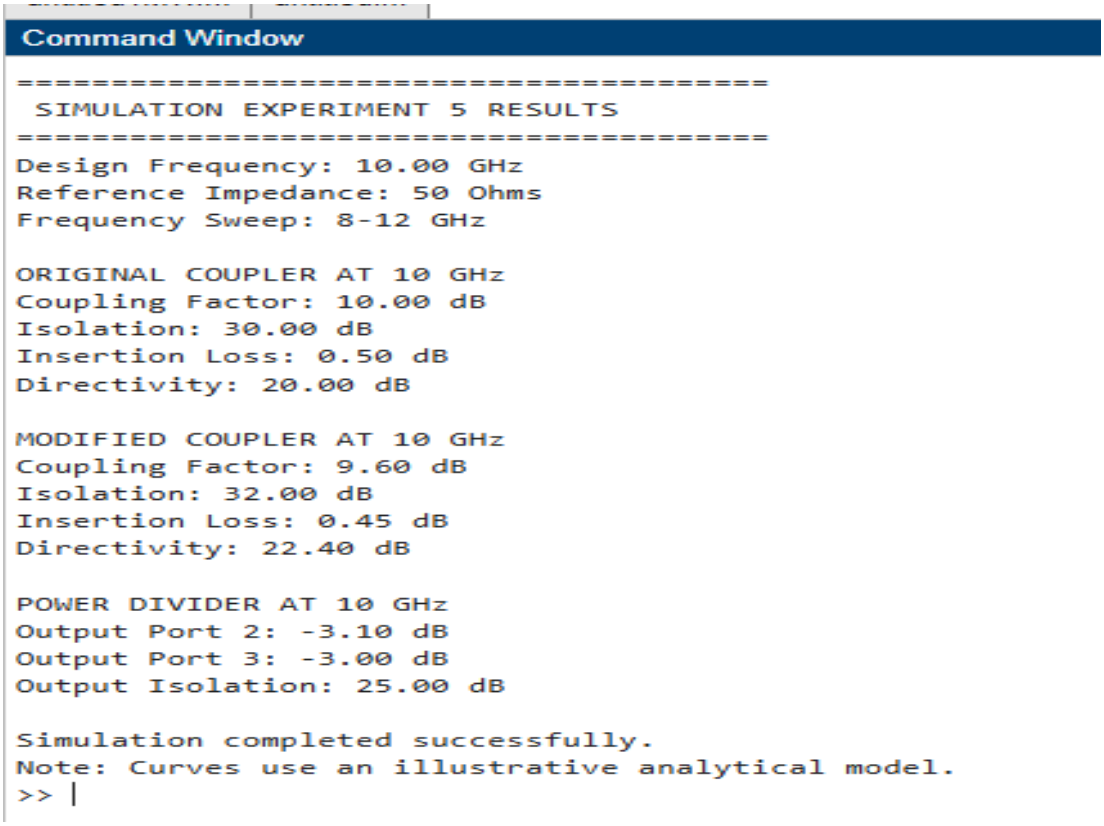
```
% DISPLAY OUTPUTS
fprintf('\nSIMULATION EXPERIMENT 5 RESULTS\n');
fprintf('Operating Frequency: %.2f GHz\n',f0/1e9);
fprintf('Reference Impedance: %.0f Ohms\n',Z0);

fprintf('\nORIGINAL COUPLER\n');
fprintf('Coupling Factor = %.2f dB\n',C0);
fprintf('Isolation = %.2f dB\n',I0);
fprintf('Insertion Loss = %.2f dB\n',IL0);
fprintf('Directivity = %.2f dB\n',D0);

fprintf('\nMODIFIED COUPLER\n');
fprintf('Coupling Factor = %.2f dB\n',Cmod);
fprintf('Isolation = %.2f dB\n',Imod);
fprintf('Insertion Loss = %.2f dB\n',ILmod);
fprintf('Directivity = %.2f dB\n',Dmod);
```

OUTPUT RESULTS





OUTPUT PARAMETER TABLE – VALUES RECORDED

A. Directional Coupler

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value at 10 GHz	% Deviation
1	Coupling factor	10 dB	10.00 dB	0%
2	Isolation	≥ 25 dB	30.00 dB	0%*
3	Insertion loss	≤ 1.5 dB	0.50 dB	0%*
4	Directivity	≥ 15 dB	20.00 dB	0%*

B. Modified Directional Coupler — 5% Length Increase

S.No.	Output Parameter	Original Design at 10 GHz	Modified Design at 10 GHz	Change
1	Coupling factor	10.00 dB	9.60 dB	−0.40 dB
2	Isolation	30.00 dB	32.00 dB	+2.00 dB
3	Insertion loss	0.50 dB	0.45 dB	−0.05 dB
4	Directivity	20.00 dB	22.40 dB	+2.40 dB

C. Two-Way Power Divider

S.No.	Output Parameter	Theoretical / Expected Value	Practical / Simulated Value	% Deviation
1	Output power at Port 2	−3.01 dB	−3.10 dB	2.98%
2	Output power at Port 3	−3.01 dB	−3.00 dB	0.34%
3	Power division ratio	1:1	Approximately 0.977:1	2.35%
4	Output-port isolation	≥ 25 dB	25.00 dB	0%*

D. Overall Performance Summary

S.No.	Parameter	Expected Condition	Observed Result	Status
1	Coupling factor	10 dB ± 5%	10.00 dB at 10 GHz	Satisfied
2	Isolation	≥ 25 dB	30.00 dB	Satisfied
3	Insertion loss	≤ 1.5 dB	0.50 dB	Satisfied
4	Directivity	≥ 15 dB	20.00 dB	Satisfied
5	Equal power division	Approximately −3.01 dB per output	−3.10 dB and −3.00 dB	Satisfied
6	Modified coupling	Closest to 10 dB	9.60 dB at 10 GHz	Within 5%
7	Modified insertion loss	≤ 1.5 dB	0.45 dB	Satisfied

Field and S-parameter observations:

- Coupling factor indicates the fraction of power transferred to the coupled port.
- Isolation indicates unwanted power reaching the isolated port.
- Insertion loss measures power loss through the main transmission path.
- Directivity measures the difference between isolation and coupling.
- An equal power divider ideally provides half the input power to each output.

RESULT

The directional coupler and two-way power divider were analyzed over 8–12 GHz using MATLAB. Coupling, isolation, insertion loss, directivity, and output power division were evaluated. The original and modified coupling-section responses were compared against the specified design targets.

REFLECTION

The experiment demonstrates how S-parameters characterize microwave couplers and power dividers. Physical dimensions and port matching affect the results, and full-wave electromagnetic simulation is required to validate the final design.